

Coding & Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/pushpa-sree/smart-Lender- https://github.com/pushpa-sree/smart-Lender
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Flask-SQLAlchemy, Scikit-learn
Key Features Implemented	User Registration & Login, Role-Based Access Control, Loan Eligibility Prediction using Random Forest, Data Preprocessing, Prediction History, Admin Dashboard, Officer Dashboard, Analyst Dashboard, CSV Export, Responsive Web Interface
Pending / Incomplete Features	None. All planned features have been successfully implemented, tested, and documented.
Setup / Run Instructions	Clone the repository, install dependencies using <code>pip install -r requirements.txt</code> , train the model (if required), and run the application using <code>python app.py</code> . Access the application at http://127.0.0.1:5000 .

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	YES
2	Meaningful variable and function names are used	YES
3	Code includes comments / documentation where necessary	YES
4	Error handling is implemented for critical operations	YES
5	The application runs without critical errors	YES
6	Code is committed to a version control repository	YES

Additional Notes / Comments:

The Smart Lender project follows a modular software engineering approach by separating machine learning, authentication, database management, and presentation logic into independent components. The application uses a trained Random Forest classifier for loan eligibility prediction and integrates secure user authentication, role-based authorization, SQLite database management, and prediction history tracking. The project is maintained using Git and GitHub, enabling collaborative development, version control, and continuous enhancement.